Partial Fractions and Improper Integrals

Section 1: Linear Partial Fractions

1 Evaluate $\int \frac{1}{x^2-4}dx$.

a. $\frac{1}{4}\ln\left|\frac{x-2}{x+2}\right| + C$

b. $\frac{1}{2}\ln\left|\frac{x-2}{x+2}\right| + C$

c. $\frac{1}{4}\ln|x^2-4| + C$

d. $\frac{1}{2}\arctan\left(\frac{x}{2}\right) + C$

2 Evaluate $\int \frac{5x-3}{(x-1)(x+3)}dx$.

a. $\frac{1}{2}\ln|x-1| + \frac{9}{2}\ln|x+3| + C$

b. $2\ln|x-1| + 3\ln|x+3| + C$

c. $\ln|x-1| + 4\ln|x+3| + C$

d. $5\ln|x^2+2x-3| + C$

3 When must long division be performed BEFORE applying partial fractions?

a. When the degree of the numerator is greater than or equal to the degree of the denominator

b. When the denominator has repeated roots

c. When the numerator is a constant

d. Always

Section 2: Repeated Linear & Irreducible Factors

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4 What is the form of the partial fraction decomposition for $\frac{x+2}{x(x-1)^2}$?

a. $\frac{A}{x}+\frac{B}{x-1}+\frac{C}{(x-1)^2}$

b. $\frac{A}{x}+\frac{B}{(x-1)^2}$

c. $\frac{A}{x}+\frac{Bx+C}{(x-1)^2}$

d. $\frac{A}{x}+\frac{B}{x-1}$

5 Evaluate $\int \frac{1}{x^2+9}dx$.

a. $\frac{1}{3}\arctan(\frac{x}{3})+C$

b. $\arctan(\frac{x}{3})+C$

c. $\frac{1}{9}\ln(x^2+9)+C$

d. $\frac{1}{3}\ln|x+3|+C$

6 Evaluate $\int_2^3 \frac{1}{x(x-1)}dx$.

a. $\ln(\frac{4}{3})$

b. $\ln 2$

c. $\ln 3$

d. $\frac{1}{6}$

Section 3: Improper Integrals with Infinite Limits (Type 1)

7 Evaluate $\int_1^\infty \frac{1}{x^2}dx$.

a. 1

b. ∞ (diverges)

c. 2

d. 0

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8 For what values of p does the improper integral $\int_1^\infty \frac{1}{x^p} dx$ converge?

a. $p > 1$

b. $p \geq 1$

c. $p < 1$

d. All real p

9 Evaluate $\int_0^\infty e^{-2x} dx$.

a. $\frac{1}{2}$

b. 1

c. ∞ (diverges)

d. 2

Section 4: Improper Integrals with Discontinuous Integrands (Type 2)

10 Evaluate the improper integral $\int_0^1 \frac{1}{\sqrt{x}} dx$.

a. 2

b. 1

c. ∞ (diverges)

d. $\frac{1}{2}$

11 Evaluate $\int_0^1 \frac{1}{x} dx$.

a. Diverges to ∞

b. 1

c. 0

d. e

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12 Evaluate $\int_{-1}^{1} \frac{1}{x^2} dx$.

a. Diverges

b. -2

c. 0

d. 2

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